Statistical Inferences About Parameters.

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Statistical Inferences

- In the previous lectures, we showed how to calculate estimates for our two linear parameters (slope and intercept).

- We also showed how to interpret these parameters, especially as the zero-point on x was changed or the scale of x was changed.

- Now, we will address how to make statistical inferences about parameters.
  - I.e., given what we observe in the sample, what can we say about the population.

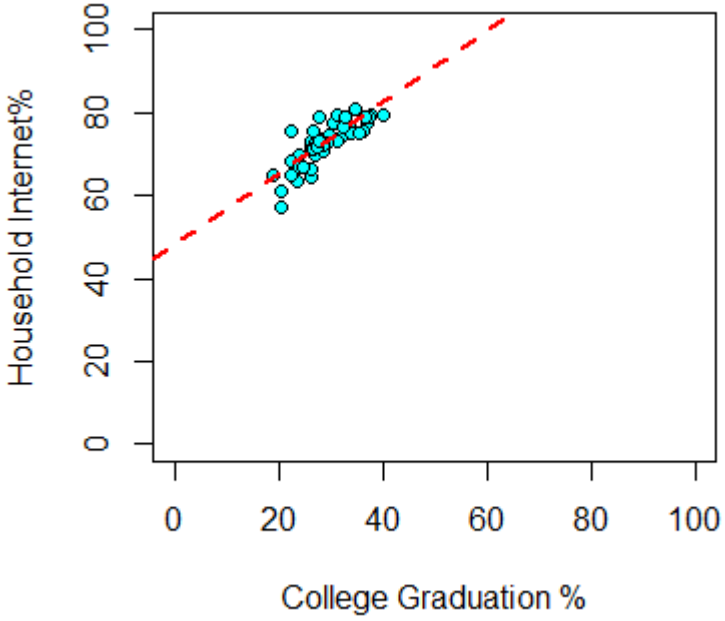
Model Comparison

Model C:

$$y_i = \beta_0$$

Model A:

$$y_i = \beta_0 + \beta_1(x_i)$$



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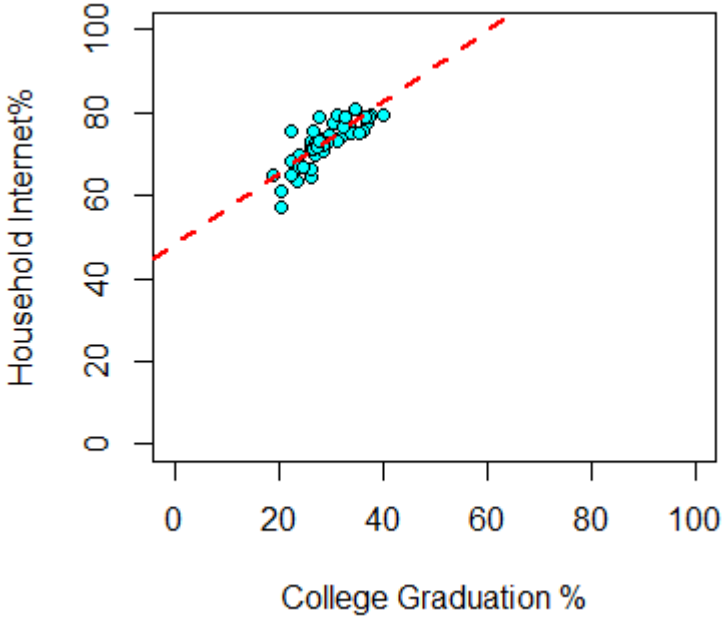
Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Intercept:

$$b_0 = \bar{y} - b_1\bar{x}$$



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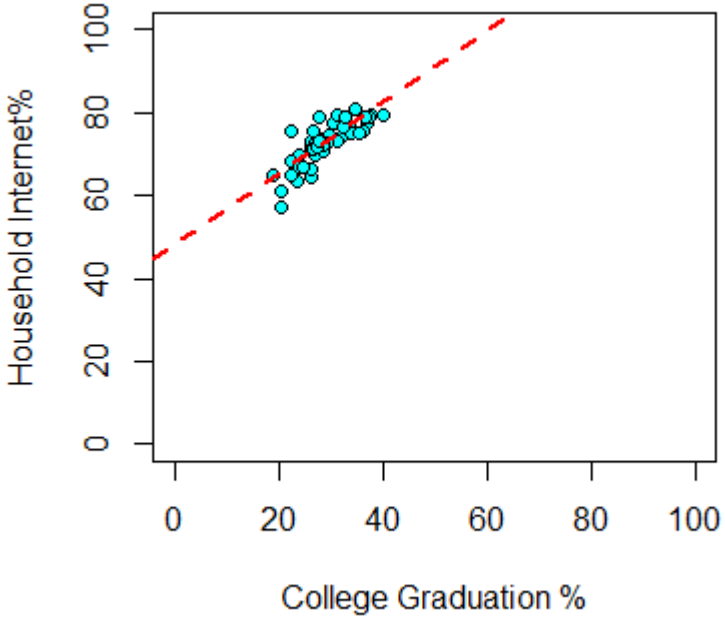


```
> head(DAT1)
  State Internet College
1    AK     63.5    23.5
2    AL     79.0    28.0
3    AR     60.9    20.6
4    AZ     73.9    27.4
5    CA     77.9    31.0
6    CO     79.4    37.8
…
```

Statistical Inferences for $\boldsymbol{\beta_1}$

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2    AL     79.0    28.0 72.806   6.194 72.0480  6.9520
3    AR     60.9    20.6 72.806 -11.906 65.6692 -4.7692
4    AZ     73.9    27.4 72.806   1.094 71.5308  2.3692
5    CA     77.9    31.0 72.806   5.094 74.6340  3.2660
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$$\boxed{\mathbf{0.6457} = \frac{1355.028 - 480.0041}{1355.028}}$$

Statistical Inferences for $\boldsymbol{\beta_1}$

- Once we have a calculation of PRE, we can calculate an F statistic.

$$F_{obs} = \frac{PRE/(PA - PC)}{(1 - PRE)/(N - PA)}$$

$$F_{obs} = \frac{0.6457/(2 - 1)}{(1 - 0.6457)/(50 - 2)} = 87.5017$$

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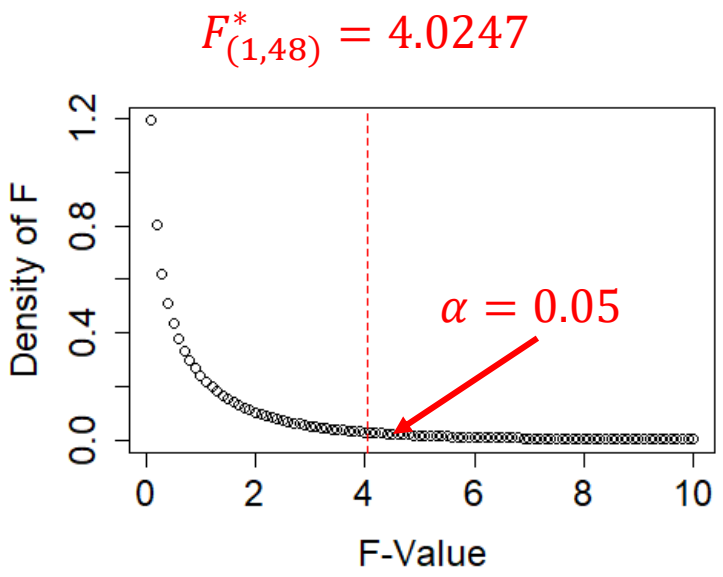
- We can then compare F_{obs} to the null distribution for $F_{(1,48)}$.



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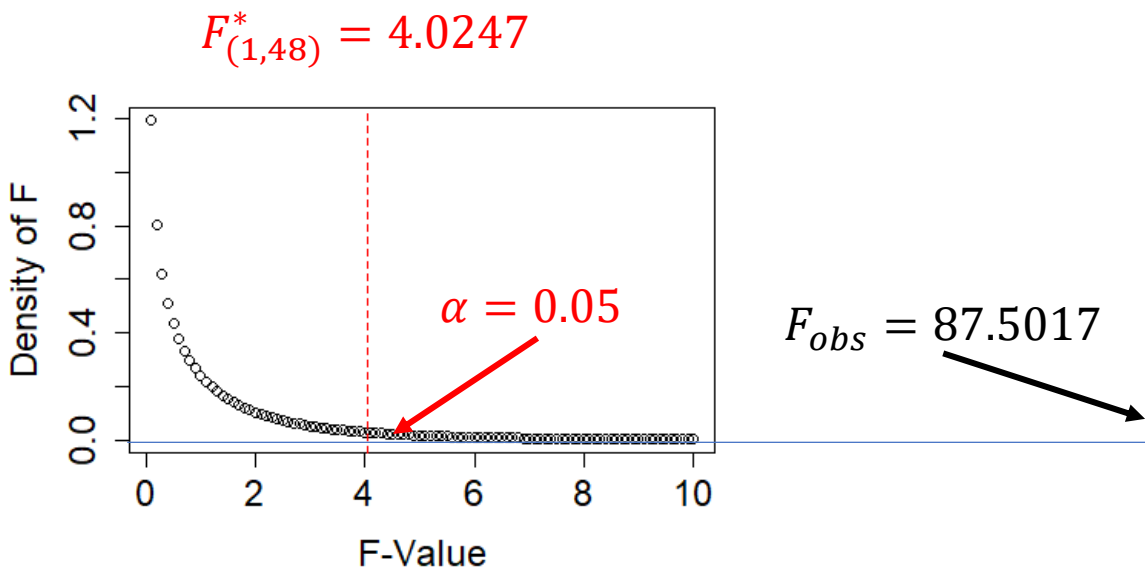

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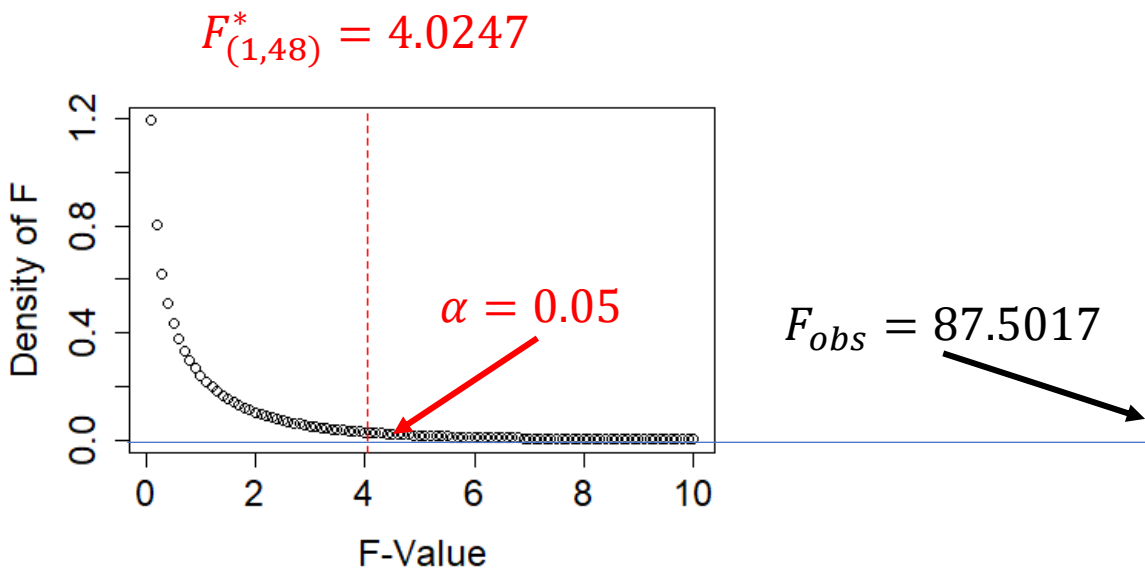

$$\boxed{F(1,48) = 87.50, p < 0.001}$$

Linear Model Output from R

```
> mod1<-lm(Internet~College, data=DAT1)
> summary(mod1)
Call:
lm(formula = Internet ~ College, data = DAT1)
Residuals:
   Min      1Q Median      3Q     Max
-8.094 -1.722 -0.175  1.470  8.296
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 47.91227    2.69854  17.755  < 2e-16 ***
College      0.86185    0.09213   9.354 2.15e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.162 on 48 degrees of freedom
Multiple R-squared:  0.6458,	Adjusted R-squared:  0.6384
F-statistic:  87.5 on 1 and 48 DF,  p-value: 2.15e-12
```

Thus far, we shown how R is calculating:

- **The point-estimate for the slope and intercept.**
- **The PRE for the augmented model.**
- **The F-observed and it's corresponding p-value.**

The books walks through:

- How the F-test for a single parameter can also be achieved with a t-test.
- How to calculate the adjusted (unbiased) R^2.

Finally, we need to understand how R is making a statistical inference about the intercept, β_0.

Statistical Inferences for $\boldsymbol{\beta_0}$

- Recall that the point estimate for $\boldsymbol{\beta_0}$ is calculated after $\boldsymbol{\beta_1}$. $b_0 = \bar{y} - b_1\bar{x}$

- This is because having a slope in our model changes the value of the intercept (predicted y when $x = 0$).

Statistical Test for the Slope

Model C: $y_i = \beta_0 + 0(x_i)$

Model A: $y_i = \beta_0 + \beta_1(x_i)$

Statistical Test for the Intercept

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Statistical Test for the Intercept

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- If you calculate the SSE, PRE, and F-observed for this model you get $F(1,48) = 315.24, p < 0.001$.
- Although R presents a t-test, these results are equivalent ($17.755^2 = 315.24$).
- And we would conclude that our observed intercept (47.91) is unlikely to have arisen under the null hypothesis ($\beta_0 = 0$).

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Is the parameter statistically different from 0?

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